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Review

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An analytical overview of the composition and characteristics of China's food safety standards

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Abstract

This paper discusses the framework of China's food safety standards and provides a brief overview of the problems and developmental characteristics of food safety in China. The composition and characteristics of China's food safety standards are revealed by an analysis of the changes in China's general food standards, an overview of the characteristics of the hygiene requirements in the production and operation process, and an introduction to food product and test method standards. In conclusion, Chinese food safety standards are still being improved, but they must also be effectively implemented and followed up in real time in order to continuously improve the quality of food and reduce food safety incidents.
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Keywords: food; standards; health codes; policy; quality

INTRODUCTION

Food safety is closely related to people's health and the stability of the food market. Therefore, it is of vital importance to ensure the quality of ingredients and products and to promote the standardization of food production and processing.[1] Currently, the EU, the USA, Japan, Singapore, Canada and other countries are increasing requirements and standards for food products. Recently, the EU introduced additional stringent hygiene and safety requirements for the chain of industries involved in the transportation and storage of food products, and has referred to private food safety standards to enrich the current system.[2] The British Food Standards Agency (BFSA), on the other hand, has set up various food standards to safeguard the interests of consumers and protect public health.[3] The USA and the EU have coordinated their standards for food safety,[4] and the USA passed the Food and Drugs Act and the Federal Food, Drug and Cosmetic Act back in 1906 and 1938, respectively, to improve food safety standards. In recent years, the United States Food and Drug Administration has also imposed higher requirements on water sources used for irrigated agriculture, strictly limiting the content of species microorganisms.[5] The spate of food safety problems in Japan since the beginning of the 21st century posed a serious danger to society and even led to a crisis of public confidence in the government and scientific experts. In response to this crisis, Japan enacted the Food Safety Basic Act in 2003 to regulate food safety issues and amended the Food Sanitation Act to clearly state that the law is intended to protect the health of consumers.[6] At the same time, countries are constantly updating their food safety policies according to their own situation, especially for developed countries like the USA, where a more complete body for making food safety policies has been formed.[7] For example, the federal government has set up the Department of Agriculture, the Department of Food and Health, the Department of Commerce, the Office of Food Administration and Budget, the Environmental Protection Agency, and other government departments with related functions, and established the 'Presidential Commission on Food Safety Management'. The USA has also adopted the HACCP (Hazard Analysis and Critical Control Point) system, a policy system that can well prevent and control food safety problems. The key points in the production, distribution and sale process are closely monitored and, once problems arise, remedial measures are taken immediately to minimize the hazards. Other countries have also started to use the system to reduce food safety problems.[8]

Researchers in South Korea discovered that milk fat and protein levels in milk from HACCP-certified dairy farms were significantly greater than in non-certified farms. The implementation of HACCP

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helped to guarantee the hygienic quality of the milk production process, resulting in higher economic incentives. These outcomes suggest that HACCP is a useful tool in ensuring the quality of food products from the source.[9] However, the USA faced many challenges when developing the food standard system in the 20th century and went through a host of failures, such as the New Diet food marketing campaigns of the 1950s and 1960s, which ultimately led to the failure of the food standards system.[10] Compared with other countries, China's food safety standards are still evolving and it has developed a framework for food safety standards with its own characteristics. Up to now, China has issued more than 1000 sets of food safety standards and related normative documents,[11] which fall into four major categories: general standards, product standards, production specification standards and inspection method standards. These four types of standards complement each other and control different food safety risks from different perspectives, covering the major food categories and health hazards in China. This paper discusses the characteristics of China's food safety policies and physique, its problems and later developments, in the hope of facilitating the establishment of a sound national food safety standard system with Chinese characteristics.

CHARACTERISTICS AND PROBLEMS OF CHINA'S FOOD SAFETY SYSTEM

At present, China has established a food safety management body with multi-departmental synergy (Fig. 1). The synergy of several departments forms a holistic management system as in the USA. In terms of the legal and regulatory system, China has enacted several laws and regulations for food safety, including *Food Safety Law of the People's Republic of Chin*a and the *National Standards for Food Safety*, which have been subsequently optimized according to national conditions.[12] In addition, China has imposed strict requirements on the indicators in each process, from growing, producing and processing to selling food raw materials. In 2013, a special food safety testing agency, the Food Safety Testing Center under the Food and Drug Administration, was established to monitor the whole process of food production. However, this testing is mostly conducted by enterprises or is based on regular sampling, and some vendors do not follow the requirements, which leads to loopholes.[13] However, for better management, food safety testing-related functions have now been subsumed under the Market Surveillance Administration. At the same time, different government departments are required to share food safety information to the community through various communication channels. The Shanghai Food and Drug Safety Commission has established the Shanghai Food Safety Network for the publication of information related to food quality, quality inspection and food safety risks.[14] Regular food safety testing results are publicized on the government website of Jiaxiang County, Jining City, Shandong Province. More information on food safety and related inspection results are released on some public media, which helps to raise public awareness of food safety.[15]

The source of food greatly influences food safety, especially the possible illegal and excessive addition of pesticides, fertilizers and feed additives, heavy metals and industrial pollution in the arable soil, water sources, feeding and breeding processes. The residual chemical substances are hard to degrade and will circulate through the food chain in the environment and linger in the human body for a long time, thus affecting human health through long-term accumulation.[16] According to statistics from the Food Safety Supervision and Sampling Inspection of the State Administration of Market Supervision and Administration of China for the first half of 2023, 70 174 batches of food products failed to meet the relevant national safety standards among the 292 1763 samples inspected. The data obtained from the sampling indicated that the majority of instances were as follows: pesticide residue levels exceeding the established standard constituted 42.98%, microbial contamination made up 14.67%, the use of food additives beyond the prescribed limits represented 13.06%, organic contamination issues accounted for 9.21%, and veterinary drugs exceeding the standard constituted 7%. The concentration of heavy metals was 7.83% in the sample, while the quality standard of 4.92% was not met. Moreover, the data indicates that the presence of pesticide residues exceeding the standard has a significant impact on food safety issues.[17] Additionally, it reveals that China still needs to address food safety concerns comprehensively. In 1999, an outbreak of *Escherichia coli* in the provinces of Jiangsu and Anhui in China led to the hospitalization of 195 people and the deaths of 177 individuals.[18] Wang and colleagues conducted an analysis of 70 samples of various types of meat from retail suppliers in Beijing, China. The results indicated that *Listeria monocytogenes* contamination was present in seven samples of frozen pork and one sample of frozen chicken.[19] As early as

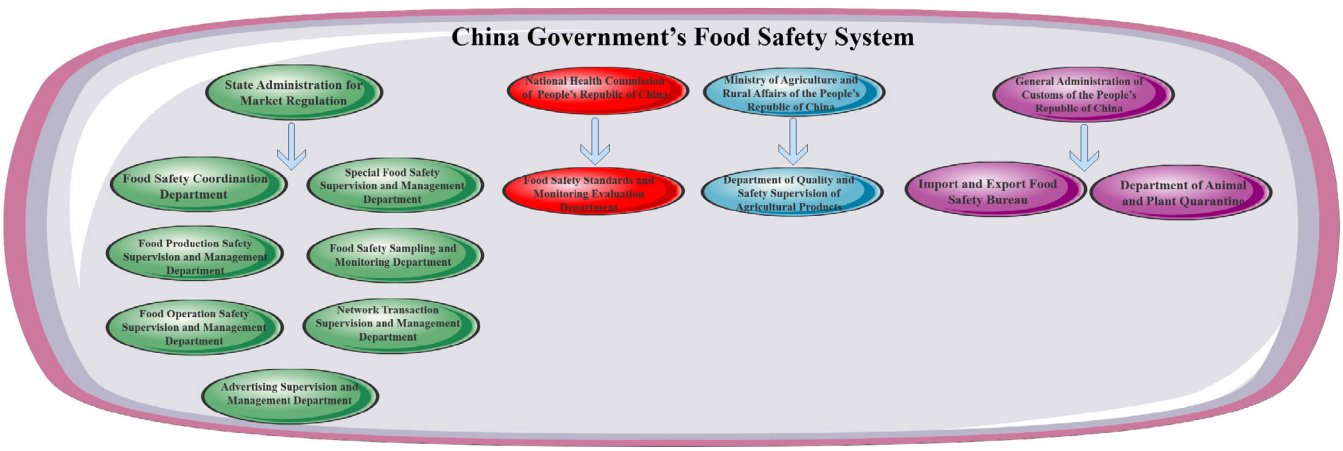


Figure 1. China Government's Food Safety System.

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2007, Professor Pan Genxing from Nanjing Agricultural University tested more than 170 rice samples from six regions in China and found that 10% of the rice sold in the market had excessive cadmium.[20] Worse still, soil and vegetables were contaminated due to excessive use of pesticides and veterinary drugs, mining pollution, animal manure and wastewater irrigation, etc.[21] Furthermore, additives like bleaching agents and dyes present on food packaging can taint food throughout the manufacturing and preservation process.[22] These problems are difficult to control.

China once had many departments in charge of food safety. When problems arose, they tended to pass the buck to each other. Later, the responsibilities of the relevant departments were redefined. Since the 20th century, food safety issues were mainly monitored by quality inspection departments and health departments, and in 2013 the Food and Drug Administration Office of the Council Food Safety Commission was restructured into the Food and Drug Administration, which is responsible for the implementation of unified supervision and management of food safety and the safety and effectiveness of drugs in production, distribution and consumption throughout the country. Since 2018, the State Administration of Market Supervision and Administration was established to replace the Food and Drug Administration, thus unifying the functions of food safety management. While the regulatory system began to taken shape, some remote areas suffered from incomplete regulation and insufficient supervision, which could easily lead to food safety incidents.[23]

In some small and medium-sized enterprises, the responsibility for food safety is not clearly implemented or not implemented at all. Businesses only care about profits, and they do not have a strong sense of responsibility for food safety, nor do they have a good understanding of the laws and regulations related to food safety, which leads to problems.[24] In particular, some private enterprises lack strict food hygiene standards, quality and safety control during processing, and do not operate strictly according to food quality regulations during production operations, resulting in food contamination. In addition, some enterprises, in order to reduce costs and pursue higher profits, use illegal ingredients, overuse food additives and produce substandard food, all of which are a result of insufficient awareness of the food safety legal system and a weak sense of responsibility. In addition, due to the wide variation in enterprise development, only a few large enterprises have their own quality standard control and testing centers, which send samples to regulatory agencies at all levels for testing. Overall, there is a significant delay in managing food safety within China's current food system, and government departments only establish appropriate standards when serious food safety issues arise. This issue demands further attention in China's current food control practices. Although this is an inevitable reality, reducing the management delay and strengthening the control of food productions, processing, and sales is of paramount importance. Furthermore, with China's enormous population, prior to 10 years ago, citizens did not have a high demand for food. Consequently, during periods of food safety awareness and low awareness of rights, food manufacturers and sellers were granted the opportunity to significantly impact food safety in China.

MANAGEMENT OF FOOD SAFETY IN DIFFERENT COUNTRIES

Food laws and standards are equally crucial in food control. Laws are mandatory, while standards are more relaxed. Differences in regulations and food safety standards exist among countries. As a result, selling agricultural products and importing/exporting related industrial chains are restricted between countries. Europe and the USA are recognized as the world's two 'superpowers' in terms of the stringency of food safety standards, and Europe is also considered 'the predominant regulator of global commerce'.[25]

Over the past few decades, a multitude of food scandals and scares have surfaced globally, including within the EU, the USA, and several other countries. Such events have eroded public trust in the safety and integrity of the food industry.[26] The most infamous scandals are the outbreak of mad cow disease, also known as bovine spongiform encephalopathy, in 1996, the animal feed dioxin crisis in 1999, avian influenza (H5N1) in 2003, the occurrence of swine feed dioxin in 2008, the outbreak of swine influenza virus (H1N1) in 2009 and the controversy surrounding China's melamine-contaminated infant formula in 2007. These food scandals necessitate stricter and more comprehensive regulation of food safety. Although the European regulations (Regulation (EC) No. 178/2002UE) and European private standards (e.g., BRC, IFS, Global Gap, FSSC22000, SQF) are considered to be the most stringent in the world, they are the subject of disagreement and complaints from developing countries, which consider that their continuous strengthening illegally prevents them from accessing the international market.[27] Europe has higher standards for food safety than the USA, especially for aflatoxin residues in food, pesticide residues and growth promoters in livestock, such as hormones and ractopamine.[28] Karemera *et al*. found that maximum residue limits (MRLs) on produce significantly inhibit trade in fruits and vegetables between the USA and the Europe, and that a 10% reduction in MRL requirements would facilitate 5% of trade if standards were harmonized, allowing the USA to export up to $473 million of fruits and vegetables to Europe each year.[29] Europe and Japan have signed Economic Partnership Agreements to maintain long-term sustainable development of agricultural import and export under the international standards.[30] Interestingly, some countries will modify their food standards to meet the needs of exporters. Brazil has modified its food-related standards to meet the needs of the EU to guarantee the export of beef. Lafisca *et al*. found that European food legislation and standards have a direct impact on Brazilian beef production as well as the development of Brazilian beef-related legislation.[31] Yang *et al*. demonstrated that Japan's strict residue limits for pesticides and veterinary drugs have played an important role in ensuring high-quality food safety and raised the bar for poultry meat imports.[32]

With the growth of industrial industries, food safety is again threatened by various types of radioactive contamination. In the immediate aftermath of the Fukushima nuclear disaster in 2011, the Japanese government issued interim food safety guidelines, including residual limits for various radioactive substances in food and sales restrictions for certain Fukushima-origin products, and revised the Food Safety Basic Act in 2012 in response to citizens' demands.[32] Countries have agreed to allow certain levels of radioactive contaminants in specific food types, and have outlined corresponding limits in Table 1.[33] This demonstrates the ongoing evolution of global food safety regulations in response to changing environmental conditions. The EU has also set up standards for monitoring new foods after they are introduced to the market by collecting data on the potential adverse effects of foods. This official recommendation for post-market food safety monitoring is based on EU Regulation 258/97.[34] The EU's long period of food safety monitoring has also benefited from the high standards required by its Codex Committee on Food Safety, which focuses

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Table 1. Countries and types of food irradiation allowed

Country (year)	Types of food allowed
Russia (1959), Ukraine (1957)	Grains
Canada (1960), Spain (1962)	Potato, onion, wheat and the above
Japan (1972), Italy (1973), Hungary (1976), South Africa (1977)	Potato, mixed dry ingredients and the above
Belgium (1980), Philippines (1981), Chile (1982), Bangladesh (1983), China (1984), Yugoslavia (1984), Brazil (1985), Denmark (1985), France (1985), Thailand (1986), USA (1986), Argentina (1987), etc.	Asparagus, garlic, onions, potatoes, strawberry, chicken, condiments, fish, fish products, frogs' legs, mangos, onion, papaya, pulses, rice, shrimp, spices, wheat, wheat products, gum arabic, herbs, shallot, tea, tomato, pineapple, apricot, sterile meats, bovine colostrum, Camembert cheese, casein, chicken, egg products, vanilla, pear, bulbs and the above
Cuba (1990), Iran (1990), India (1991), Czech Republic (1992), Netherlands (1992), Croatia (1994), Mexico (1995), Pakistan (1996), etc.	Soy sauce powder, red pepper paste, starch and the above

on the compatibility of these standards with the EU's highest food safety objectives when adopting food-related legislation, and this has ensured the sustainable operation of the EU's stringent food safety standards.[35] The USA has stringent regulations for food-related standards and, among the contaminants in the food chain, irrigation water plays a crucial role. The USA has developed the Produce Safety Rule under the Food Safety Modernization Act. This rule mandates that all water sources used for irrigation or that may make contact with food raw materials meet specific standards. These standards require a geometric mean of ≤126 CFU 100 mL^{-1} (≤2.1 log CFU 100 mL^{-1}) and a statistical threshold value of ≤410 CFU 100 mL^{-1} (≤2.6 log CFU 100 mL^{-1}) of generic *E. coli*.[36]

CHINA'S FOOD SAFETY STANDARDS AND REGULATORY DOCUMENTS

At present, China has thousands of food safety standards, which can be roughly divided into general food standards, product standards, food processing, production and operation standards, and food inspection and testing standards. From requirements for microorganisms, pesticide and veterinary drug residues in food raw materials to standards for hygiene and additive use during production and processing, and to standards for testing physical and chemical indicators in some areas, China's food standards have been gradually improving (Fig. 2).

General food standards

'General food standards' refer to the minimum requirements for all foods, covering the basic elements in food processing and production (Table 2), such as the content of contaminants, pathogenic bacteria, mycotoxin content in food, the use of food additives, the use of nutritional fortification, pesticide and veterinary drug residues, food labeling specifications and food-grade materials. The relevant standards for residues of contaminants in food set strict requirements for the limits of residual chemical substances in food ingredients throughout the process of growing/rearing, processing/packaging, storage/transport and distribution. In particular, maximum residue limits are set for lead, cadmium, mercury, arsenic, tin, nickel, chromium, nitrite, nitrate, benzo[*a*]pyrene, *N*-dimethylnitrosamine, polychlorinated biphenyls and 3-chloro-1,2-propanediol in foodstuffs.[37] This standard was revised in 2022 in response to the latest international requirements and the demand for adequate contamination data on dry products. The contaminant content was further limited and the judgment process was simplified. The standard for lead content in food became more stringent. With the accumulation of lead contamination data on Chinese food, the new standard focuses on the lead limit for food for infants and children, liquid milk, fruit juice, honey, vegetable products and fruit products with reference to the requirements of the International Codex Alimentarius Commission (CAC) (Table 3). This facilitates the implementation of the requirements of the Joint FAO/WHO Expert Committee on Food Additives (JECFA) to reduce the content of lead in food and thus protect the health and safety of the public.[38] In addition, this revision also involved modifications on food-specific terminology, food categories and certain characteristics of heavy metal residues such as cadmium, mercury and arsenic. This newly revised standard came into effect on June 30, 2023. In terms of the pathogenic bacteria limits and detection methods for pre-packaged food and bulk food, China has developed two standards: GB29921-2022 and GB31607–2022. The precise limits of pathogenic bacteria, such as *Staphylococcus aureus*, *Salmonella*, *L. monocytogenes*, *E. coli* (diarrhea-causing), *Vibrio parahaemolyticus* and *Cronobacter* spp. (*Enterobacter sakazakii*), have been defined for different types of food. For the limits of mycotoxins in food, China has developed GB 2761-2017 based on GB 2761-2011 according to the actual situation. The limits for aflatoxin B1, aflatoxin M1, deoxynivalenol, penicillin, ochratoxin A and zearalenone in foodstuffs have been revised. Compared to GB 2761-2011, GB 2761-2017 has been modified and supplemented, mainly concerning the content limits of ochratoxin A in wine and coffee and that of contaminants in special dietary products. Mycotoxins are an important factor endangering food safety, such as canned foods, pasta products, dairy products and sauces. Mycotoxins, especially aflatoxin, are common fungal toxins that are harmful to humans.[39]

Pesticide and veterinary drug residues are important contaminants for food, endangering human health, especially in rural areas of China, where the use of pesticides and veterinary drugs often exceeds the standards; thus China faces challenges in the control of pesticide and veterinary drug residues.[40,41] Currently, about 10.18% of China's farmland is contaminated with heavy metals, damaging 13.86% of food production, and long-term intake of food containing heavy metal contamination can cause serious health hazards.[42] As for pesticide and veterinary drug residues in food, China has developed GB 2763-202 and GB 31650-2019, which cover 564 pesticides and 267 veterinary drugs, including 428 kinds of registered pesticides and 49 kinds of prohibited pesticides. The standard also sets MRLs for 104 veterinary medicinal products (classes) used in animal feed and lists 154 veterinary medicinal products that may be used to treat animal diseases but for which no MRL is required. In addition, there are nine veterinary medicinal products which may be used to treat

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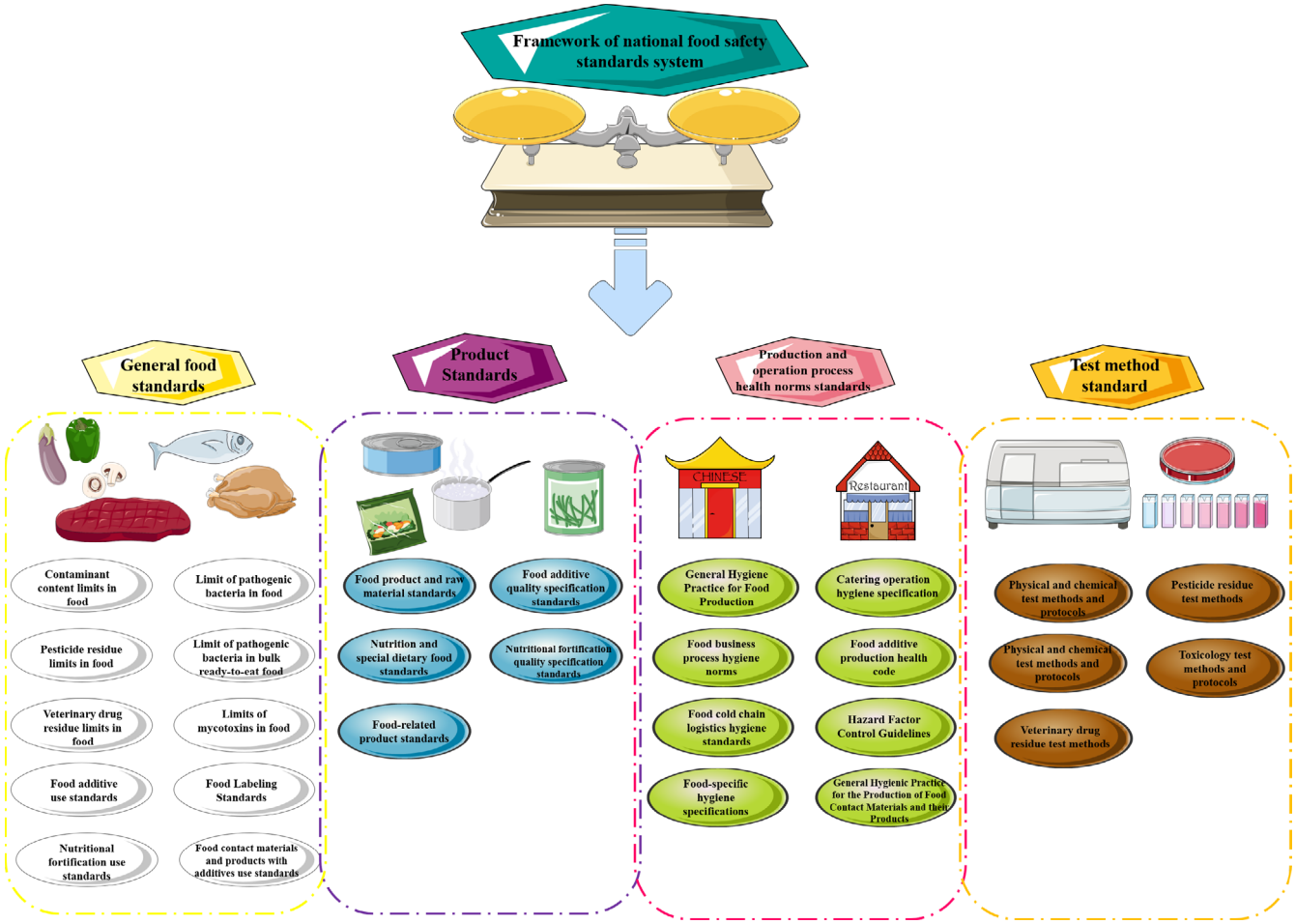


Figure 2. Framework of China's food safety standards system.

Table 2. General standards in food raw materials

Standard types	Number of contents involved	Limited quantity	Standard code	Time of implementation
Limits of contaminants in food	13 types of pollutants	357	GB 2762-2017; GB 2762-2022	2017-09-17 2023-06-30
Limits of pathogenic bacteria in food	11 species of pathogenic bacteria	185	GB29921-2022; GB31607-2022	2021-11-22 2022-03-07
Limits of mycotoxins in food	6 species of fungi	59	GB 2761-2017	2017-09-17
Food additive use standards	2374 types of additives	4 280	GB 2760-2014	2015-05-24
Nutritional fortification use standards	164 types of nutritional fortifications	320	GB 14880-2012	2013-01-01
Limits of pesticide residue in food	564 types of pesticides	10 092	GB 2763-2021	2021-09-03
Limits of veterinary drug residue in food	267 types of veterinary drugs	2 191	GB 31650-2019	2020-04-01

animal diseases but which may not be detected in food. In September 2022, the Ministry of Agriculture and Rural Affairs, the National Health and Wellness Commission and the General Administration of Market Regulation jointly issued a new national standard for food safety – GB 31650.1-2022 – which is to be used in conjunction with GB 31650-2019 as an addendum to the original standard and came into effect on February 1, 2023. However, the consumption of pesticides and veterinary drugs in China has been on an upward trend over the past 20 years, reaching an intensity of 13.07 kg ha^{-1} as of 2019 – roughly four times that of the UK and as much as five times that of the USA. According to the State Administration of Market Supervision, 36.42% of substandard food products are due to excessive pesticide residues.[43] Sun *et al.* predicted that cities such as Henan, Anhui, Hebei and Chongqing will face high risks of agricultural pollution if remedy measures are not taken.[44]

Some naturally derived or chemically synthesized substances are often added to foods to improve certain physicochemical properties, such as extending shelf life, flavor modification, and stability enhancement.[45,46] However, excessive addition of such additives can increase harm to humans. In response to this, China has formulated GB 2760-2014 for food additive limit

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Table 3. Changes in food contaminant residue standards before and after revision

Types of food involved	Modifications
Vegetables and their products	The lead limit for pickled vegetables, dried vegetables and other vegetable products is adjusted to 0.5, 0.8, and 0.3 mg kg^{-1} respectively.
Fruits and their products	The lead limit for jams (purees), preserves, dried fruits and other fruit products is adjusted to 0.4, 0.8, 0.5 and 0.2 mg kg^{-1}, respectively, with reference to the relevant CAC standards.
Edible mushrooms and their products	The lead limit is revised to 0.3 mg kg^{-1} for double spore mushrooms, flat mushrooms, shiitake mushrooms, hazel mushrooms, 1.0 mg kg^{-1} for porcini mushrooms, pine mushrooms, truffles, greenhead mushrooms, chanterelles, chanterelles, juicy milk mushrooms. The lead limit for wood ear and its products is set at 1.0 mg kg^{-1} according to the dry weight and that for other edible mushrooms and their products is revised to 0.5 mg kg^{-1}.
Beans and their products	The lead limit for soybean products (except soy milk) is adjusted from 0.5 to 0.3 mg kg^{-1}.
Algae and their products	The lead limit for fresh algae (except spirulina), algae products (except spirulina products) is adjusted to 0.5, 1.0 mg kg^{-1}, regardless of dry weight.
Nuts and seeds	The lead limit for raw coffee beans and roasted coffee beans remains unchanged (0.5 mg kg^{-1}).
Meat and meat products	The lead limit for offal products is set at 0.5 mg kg^{-1} and that for other meat products is adjusted to 0.3 mg kg^{-1}. The lead limit for dried meat products should be converted with the dehydration rate.
Aquatic products	The lead limit for fish products is adjusted from 1.0 to 0.5 mg kg^{-1}.
Milk and dairy products	The lead limit for raw milk, pasteurized milk and sterilized milk is lowered from 0.05 to 0.02 mg kg^{-1}. The lead limit in prepared and fermented milks reduced to 0.04 mg kg^{-1}. The lead limit for other dairy products such as milk powder is revised to 0.2 mg kg^{-1}.
Preserved egg, preserved egg sausage	The lead limit for eggs and preserved egg is adjusted from 0.5 to 0.2 mg kg^{-1}.
Oil and its products	The lead limit in oils and its products is adjusted from 0.1 to 0.08 mg kg^{-1}.
Condiments	The lead limit for edible salt is adjusted from 2.0 to 1.0 mg kg^{-1}. The lead limit for pepper, cinnamon and a variety of mixed spices is maintained at 3.0 mg kg^{-1}. The lead limit for other spices and their products is revised to 1.5 mg kg^{-1}.
Liquor	The lead limit for distilled spirits (other than white wine) is adjusted to 0.2 mg kg^{-1}.
Food for special dietary uses	The lead limit for powdered infant formula is revised from 0.15 to 0.08 mg kg^{-1}, and that for complementary foods for infants to 0.2 mg kg^{-1}.
Other categories	The lead limit for jelly and honey is revised from 0.5 to 0.4 mg kg^{-1}, from 1.0 to 0.5 mg kg^{-1}, respectively. The lead limit for rape pollen, pine pollen and other pollen is adjusted to 1.0, 1.5, 0.5 mg kg^{-1}, respectively.

requirements and is proposing new requirements, which have called for public comments since September 2022. As for the use of nutritional fortification, China also developed GB 14880-2012. The national standard for food additives GB 2760-2014 regulates the upper limit of additives such as colorants, flavor enhancers, emulsifiers, anti-caking agents, coating agents, preservatives and sweeteners in food. Especially in recent years, due to easier access to media communication, Chinese people have increasing resistance to food additives and are reluctant to buy foods containing large amounts of additives; therefore, it is important to ensure the proper use of food additives and their production.[47] As necessary additives to foods for infants and young children, middle-aged and elderly people, and in certain efficacious foods, the use of nutritional fortification should be regulated.[48,49] Therefore, China has set strict requirements on the addition of vitamins, niacin, folic acid, pantothenic acid, inositol, biotin, choline, amino acids, functional peptides and minerals. In addition, as for food labels, there are corresponding standards on the materials and product additives in contact with food to further rationalize food production and processing.

Hygienic standards during the production and operation process

Food production and operation also lead to food safety issues; food contamination can occur in processing plants, vending stores and transportation channels, and from personnel.[50,51] Therefore, in addition to the basic general standards, China has set up 34 sets of regulatory requirements for the food production, operation, transportation and vending. On the production side, China has developed the General Hygienic Practice for Food Production GB14881-2013, the General Hygienic Practice for the Production of Food Contact Materials and their Products GB 31603-2015 and the Hygienic Practice for the Production of Food Additives, which set hygienic requirements for food, food additives and food contact materials and products, mainly concerning the production, collection, processing, packaging, storage and transportation of raw materials, and other aspects of the premises, equipment and personnel hygiene. For process assurance in operation and post-processing, there are two sets of national standards – the Food Operation Process Hygiene Code GB 31621-2014 and the General Hygiene Code for Food Service GB 31654-2021 – as well as one guidance document – Food Safety Code of Practice for Food Service (2018 Edition). China has also released hazard guidelines for safety and hygiene-related factor control. National standards are mandatory, serving as the bottom line of food safety, with corresponding legal punishment, while the guiding normative documents focus more on specific measures to ensure food safety, which are more practical and operational. In recent years, China's pre-prepared vegetables and frozen products industry has been prospering. In order to ensure hygiene during transportation, product handover, distribution, storage, personnel management and control, the Food Cold Chain

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Logistics Hygiene Specification GB 31605-2020 was formulated to specify the traceability and recall of problematic products and document management during food cold chain transportation. In addition to the above-mentioned standards and regulatory documents, 26 food-specific hygienic norms have been created for different types of food, such as frozen food, infant formula, aquatic products, special medical use formula, beverages and dairy products, so as to further ensure food hygiene in the production and operation process. In short, food hygiene in production and operation is of great significance to ensure food safety.

Product standards and test method standards

There are a wide range of food products and correspondingly, therefore, different types of food standards, which can be classified by the physical state, processing method, type of food, nutrient content, and physical and chemical properties.[52] In order to further refine the production process and quality testing standards for food products, China has developed 809 product standards, including 70 standards for food products and raw materials, 10 for nutrition and dietary foods, 651 for food additive quality specifications, 62 for nutrition fortification quality specifications, and 10 for food-related products. The development of these standards imposes stricter nutritional quality requirements in food products during production and processing.

For factors affecting food safety and quality, including physical and chemical properties, microbial counts, and pesticide and veterinary drug residues, there is also a need to establish standards for different detection methods.[53,54] China has developed testing method standards, including 327 physical and chemical testing method protocols, 32 microbiological testing methods and protocols, 29 toxicological testing methods and protocols, 120 pesticide residue testing methods, and 74 veterinary drug residue testing methods. By continuously refining the standards, we can effectively strengthen the control of food safety, so that when problems arise we can find the applicable standards in the first place to avoid the loopholes.

DISCUSSION ON THE DEVELOPMENT OF CHINA'S FOOD SAFETY STANDARDS SYSTEM

While China's food safety standards system is improving and its scientific rationality is increasing, we need to enhance supervision and management of food production, processing, transportation and sales, as well as the management of regulatory agencies and enterprise production. The regulations should significantly increase the production cost of enterprises. When the penalties imposed on companies for food problems can affect their reputations and livelihoods, food safety can be guaranteed to the greatest extent possible. Enterprises establish a food safety supervision and inspection agency as an advisory body to ensure the vertical management of the enterprise, which is given the right to supervise, inspect and revoke. When food safety problems arise within the enterprise, the agency reports to the supreme authority, which gives full authority to the agency to investigate, handle and rectify the situation. It also imposes severe penalties on individuals or institutions within the enterprise that try to block its rights, so that food safety is guaranteed within the enterprise and its integrity is enhanced. A researcher conducted a questionnaire survey of key food quality production managers in 15 companies in Saudi Arabia and 28 companies in Belgium and found that food certification standards had a greater impact on the integrity of food quality. For example, if two food companies producing similar products implement two production standards – ISO 9000 and FSSC 22000 – the quality of their products will be significantly different.[55] Therefore, the development and implementation of different food safety standards has an important impact on ensuring food quality and food integrity. In addition, there is a relationship between the type of food produced and the food integrity environment, with plant-based food companies having a higher level of food integrity than animal-based food companies. In addition, street food vendors' attitudes and practices regarding food safety should be improved, and appropriate publicity and training undertaken to promote the full implementation of food safety standards, which has implications for the development of food in countries such as India, where the street food industry is well developed.[56] There is as lack of appropriate standards for some traditional foods with local characteristics. The EU issued EC 509/2006 as early as 2006, which defined the scope of food that may be recognized as a guarantee of traditional characteristics and the traditional characteristics of food. The term 'distinctive' refers to a characteristic or group of characteristics that clearly distinguish an agricultural product or food from other similar products in the same category (a characteristic relates to the intrinsic characteristics of a product, such as its physical, chemical, microbiological or organoleptic characteristics, or to the method of production or to specific conditions in the production process, rather than to its external characteristics). According to the 2014 EU regulations, such foods should be at least 30 years old to be considered a traditional speciality food.[57] In China, existing national food safety standards do not cover all food products, and some local specialities still require additional local standards or group standards to be developed by local governments.[58] There are still foods that are difficult to distinguish accurately, thus preventing the application of precise food safety standards, which need to be further clarified and classified in the future.

The food safety regulations and standards of the EU are presently some of the most outstanding worldwide, particularly in the domains of hygiene and food safety.[25] One issue hindering food safety in the EU is the accurate comprehension and implementation of regulations by both the food industry and consumers.[59] To address this, the EU Food Safety Commission is constantly refining and updating its guidelines to further minimize food safety incidents. The issue in China is comparable, where Chinese consumers struggle to comprehend food safety policies and standards fully. Some food processors exploit customers' inadequate understanding of food safety standards for their advantage, necessitating China's food quality and safety control agencies to enforce stricter quality management. Currently, in comparison to over a decade ago, there has been some improvement in the quality of food control in China, but ample scope for further development remains. To enhance food safety control, China ought to gain further knowledge from the EU and amend food-related standards and regulations. Similarly, American food safety standards and management systems are more specific and comprehensive, and food testing and supervision are very strict.[60] Unlike China's more unified regulatory system, the USA employs a combination of decentralized and centralized regulatory models. Each government department possesses full food safety enforcement powers and regulates food safety incidents comprehensively, which encourages interdepartmental cooperation and eases the management of all aspects of the food

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industry in the country. In the past few years, there have been many problems in China's food supervision, among which the division of labor among various government departments for law enforcement supervision and management was not clear; different departments enforced the law individually, and the implementation standards were different, resulting in food safety problems not being properly handled and causing many negative effects. In recent times, the Chinese authorities have acknowledged this predicament and have started streamlining the duties of government agencies, thereby boosting the operational efficacy of the food sector. While the Chinese government has made tangible strides in delineating responsibility for food, there remains a delay in advancing and refreshing food standards, necessitating further optimization. In a word, to ensure food quality and reduce food problems, we must establish a framework for food safety standards, and update the standards in real time, so as to ensure the green and healthy development of the food industry.

CONCLUSIONS

China's national food safety standards system is gradually improving, and the standard-setting and revision process now focus more on the organic linkage between standards and food safety risk monitoring and assessment work. Food safety risk monitoring data are used to develop revised standards for important health hazards such as contaminants in food. However, it is important to ensure the effective implementation of the standards to close the food safety quality gap between China and developed countries such as Europe and the USA.

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Research data are not shared.

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